



STREETLIGHT

INSIGHT

Truck Volume Methodology and Validation

September 2022



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Introduction

Transportation professionals rely on commercial vehicle volume metrics for a wide range of projects, including pavement and bridge design and management, roadway capacity planning, environmental impact analysis, and infrastructure investment decisions. Traditional data collection methods, such as permanent and temporary traffic counters, can be expensive and challenging, have coverage limitations, and often have data quality issues. StreetLight Data, Inc. (“StreetLight”) has developed Truck Volume Metrics for the U.S. that provide average daily traffic estimates for all roads. This white paper summarizes the methodology and presents validation results for StreetLight’s Truck Volume Metrics, as developed for the years 2019 to 2021.

This document is updated regularly, so please check our website or reach out to your StreetLight contact to ensure you have the most up-to-date version of this document.

Methodology

Data Sources

The StreetLight Truck Volume models use the following sources of input data:

- **GPS trips:** Our algorithms create trips by processing location data points from commercial vehicles. Vehicle classes are derived from information about data providers and trip characteristics. The resulting sample contains trips for light-, medium-, and heavy-duty vehicles.
- **Location-based services (LBS) trips:** Our processing algorithms link pings from LBS devices into trips and then classify these trips into travel modes. The Truck Volume models use trips that we have identified to be from vehicles.
- **OpenStreetMap (OSM) data:** OSM provides information on road classification, number of lanes, speed limit, and other attributes.
- **U.S. Census data:** We use demographic data, such as population, housing units, income, and whether a location is in an urban area.
- **Weather data:** Precipitation and temperature provide another important input for the models.

Vehicle Classification

Our Truck Volume models follow a similar classification to the Federal Highway Administration (FHWA), which classifies vehicles into 13 categories (see Figure 1). We further group those

categories into light-, medium-, and heavy-duty. Consistent with FHWA, StreetLight has defined light-duty (LD) as FHWA categories 1–3, which includes passenger vehicles as well as four-tire pickup trucks and vans. StreetLight’s medium-duty (MD) encompasses categories 4–6 and includes buses and trucks of up to three axles, while heavy-duty (HD) categories 7–13, includes trucks with four or more axles. StreetLight’s Truck Volume Metrics are available for MD and HD vehicles.

Note that StreetLight’s All Vehicles Volume Metrics cover all vehicle categories, including cars and trucks.

Light Duty	Class 1 Motorcycles		Class 7 Four or more axle, single unit	
	Class 2 Passenger cars			
				
			Class 8 Four or less axle, single trailer	
				
				
Medium Duty	Class 3 Four tire, single unit		Class 9 5-Axle tractor semitrailer	
				
				
	Class 4 Buses		Class 10 Six or more axle, single trailer	
				
			Class 11 Five or less axle, multi trailer	
	Class 5 Two axle, six tire, single unit			
			Class 12 Six axle, multi-trailer	
				
	Class 6 Three axle, single unit		Class 13 Seven or more axle, multi-trailer	
				
				
				

Figure 1: StreetLight’s vehicle categories are based on the FHWA classification scheme: light-duty (LD) includes categories 1-3, medium-duty (MD) encompasses categories 4-6, and heavy-duty (HD) are categories 7-13. (Source: FHWA)

Model Summary

StreetLight’s Truck Volume Metrics are based on five machine-learning models, which link together to predict total vehicles on a road (all vehicles), as well as each vehicle class—LD, MD, and HD. Although we do not have an isolated LD Volume Metric in the StreetLight InSight®

platform, estimated LD volume is an important step in calculating volume for other vehicle categories. In this whitepaper, we focus on three classification models as the other two models calculate the All-Vehicle Volume addressed in [StreetLight's All Vehicles whitepaper](#). The three truck models use Extreme Gradient Boosting (XGBoost) algorithms and rely on similar input features. An important model constraint is that the sum of volumes for the three vehicle categories equals our All-Vehicles Volume Metric, which is one of the reasons our All-Vehicles Volume estimates are an input to the Truck Volume model.

StreetLight delivers Truck Volume across a wide range of day parts and day types within the 2019-to-2021 time span. As with our other Volume Metrics, to support volume estimations over the range of time periods, we first estimate Monthly Average Daily Traffic (MADT). That value is then scaled to the days or parts of the day requested by a particular analysis. In the scaling process, we assume the ratio between sample trip counts for a particular day or day part and trip counts for the entire month is the same as the ratio between volume for a particular day type or day part and MADT for that zone.

Validation

Ground-Truth Data

We evaluate the performance of the Truck Volume models by comparing estimated volume with volume reported by permanent traffic counters. The permanent counter data comes from the FHWA Travel Monitoring Analysis System (TMAS) CLS (vehicle classification) dataset, which includes traffic counts from more than 3,000 unique sites from January 2019 through December 2021. Data is available from 34 states in 2019 and 42 states in 2020 and 2021. Figure 2 below shows a map of permanent counter sites. (For a table of the number of counters and site locations by state, see the Appendix.) While not all states are included, the sites provide comprehensive representation of regions, road types, and urban versus rural areas across the country.

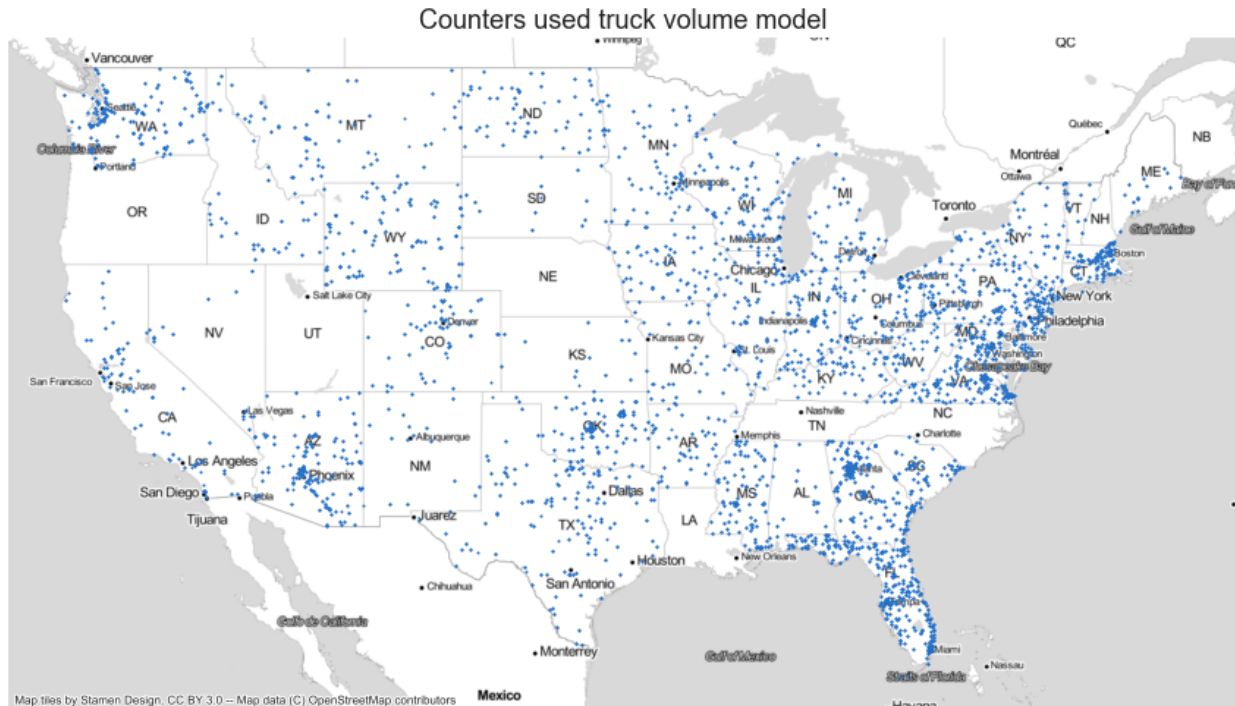


Figure 2: Locations of permanent counters used in StreetLight's Truck Volume models, 2019-2021.

Each permanent counter site in the TMAS dataset reports counts for the 13 FHWA vehicle categories, which we then group into LD, MD, and HD classes. While each state maintains different counter technology, broadly, vehicles are classified using sensors that detect the number and spacing of axles.¹ Classification errors may occur for various reasons, for example, a vehicle changing lanes, a malfunctioning sensor, or rough pavement. Therefore, our ground-truth data may include a small amount of classification error.

The distribution of permanent counters by road size is shown in Table 1, along with average daily truck traffic reported by those counters. Road size bin is based on total average daily traffic. As expected, we can see that a large proportion of counters are on high-volume roads, such as highways, with more than 10,000 in total MADT. And relatively few counters are on low-volume or rural roads with less than 500 total MADT.

Total MADT Bin	Number of counters*	Average MD MADT	Average HD MADT	MD Percent of Total MADT	HD Percent of Total MADT
A: 0–499	129	20	31	6%	9%
B: 500 –1,999	759	69	115	6%	9%
C: 2,000–4,999	1,030	168	287	5%	9%
D: 5,000–9,999	949	324	681	4%	9%
E: 10,000–19,999	1,035	620	1495	4%	10%
F: 20,000–34,999	788	1062	2733	4%	10%
G: 35,000–54,999	439	1675	3924	4%	9%
H:55,000–84,999	275	2411	4713	4%	7%
I: 85,000–124,999	145	3594	6007	4%	6%
J: 125,000+	81	5719	8483	3%	5%

*Each counter may have multiple travel directions

Table 1: Distribution of permanent counters by road size for all vehicles, where road size is based on total average daily traffic, with average MD and HD daily traffic reported by those counters.

Table 1 illustrates that trucks typically make up a small proportion of vehicles on any road. For example, on counter sites with more than 10,000 total MADT, an average 8% of vehicles are heavy-duty trucks. When reporting error metrics in the remainder of this document, we will group roads *based on the daily traffic of the vehicle class in question*—thus a highway with 50,000 total MADT may have only 3,000 HD trucks and fall into the “C:2,000–4,999” bin when grouped by HD MADT.

Measuring Accuracy

When evaluating model accuracy, we group permanent counter sites by MADT of the relevant vehicle class(es) and report results within those groupings. Errors (measured as a percentage of MADT) naturally increase with lower volume, so groupings by vehicle class-based MADT allow for more visibility into where the errors of the model lie. For each road size grouping, we report on five measures of error, each of which illuminates a different aspect of model performance, as described in Box 1 in appendix.

Because day-of-week and time-of-day Volume Metrics are derived from MADT estimates, we focus on MADT when evaluating accuracy. It’s important to note that Volume Metrics for time

¹ FHWA (2016). Traffic Monitoring Guide. United States. Federal Highway Administration.
https://www.fhwa.dot.gov/policyinformation/tmguidetmg_fhwa_pl_13_015.pdf.

periods other than month may introduce additional error.

Accuracy Benchmarks

To evaluate the performance of our models, we would ideally compare our models' accuracy metrics against those of alternative estimation methods.

Note: The FHWA does not define accuracy benchmarks for truck vehicle volumes and guidance for commercial vehicles is limited.

In the absence of industry-wide standards, we define benchmarks for our Truck Volume Metric by starting with estimated errors associated with the all-vehicles traffic estimation, and then adding an allowance to account for error in vehicle classification.

Our benchmark definitions are based on the principle that the StreetLight Truck Volume Metrics should perform at least as well as alternative methodologies. As described by the FHWA, traditional methods of estimating truck traffic rely on vehicle counts from either permanent or temporary vehicle counters, depending on the context.² Permanent (also called continuous) counters provide data for each hour of each day, so data are available for all time periods, but each state maintains only a small set of permanent counter locations, so road type factors must be applied to adjust counts to other roads. Temporary counters can be used on a much larger set of roads, but data are typically collected for a short time period (e.g., 24 or 48 hours) only every few years, and some states collect data on a given roadway only every six years. Therefore, temporary counts must be expanded using a set of factors to adjust for seasonality, growth, time of day, and/or day of week.

Each of these spatial and temporal adjustment factors add error, and when multiple factors are applied, errors compound. A study for the FHWA estimated the error associated with expansion from temporary counts taken in the same year, as shown in Table 2.³ Another analysis for the FHWA estimated typical errors in situations in which same-year counts were not available ("no count available" in Table 2).⁴ While these studies were specifically concerned with Annual Average Daily Traffic estimation, we can assume errors for Monthly Average Daily Traffic would be similar.

² FHWA (2016). Traffic Monitoring Guide. United States. Federal Highway Administration.
https://www.fhwa.dot.gov/policyinformation/tmguidetmg_fhwa_pl_13_015.pdf

³ Krile, R., Todt, F., Schroeder, J. (2015). Assessing Roadway Traffic Count Duration and Frequency Impacts on Annual Average Daily Traffic Estimation (FHWA-PL-16-008). United States. Federal Highway Administration.

⁴ Clarke, N., Co, S., Schewel, L., Wergin, J., Willoughby, C., Yan, L. (2021). "Guidelines for Obtaining AADT Estimates from Non-Traditional Sources," Non-Traditional Methods to Obtain Annual Average Daily Traffic (AADT) (FHWA-PL-21-031). United States. Federal Highway Administration.

	Temporary count expansion ⁵		No count available ⁶	
Total Average Daily Traffic	MAPE (%)	NRMSE (%)	MAPE (%)	NRMSE (%)
A: 0499	n/a	n/a	n/a	n/a
B: 500–1,999	10.0	12.9	50	65
C: 2000–4,999	10.4	17.2	50	65
D: 5,000–9,999	9.2	13.9	18	27
E: 10,000+	8.9	12.9	18	27

Table 2: Typical errors expected by alternative methodologies of estimating all-vehicles AADT, as reported by the FHWA.

In addition to these spatial and temporal adjustment factors, assumptions must be made to obtain counts by vehicle category. Some counter sites classify vehicles using sensor data on axles and vehicle weight, in which case a classification error of up to 10% may be expected.⁷ If the counter site does not provide vehicle class, then classification must be done via adjustment factors, which add error that is presumably greater than 10%. In our benchmarks, we assume vehicle classification error adds a 10% error to the errors associated with the All-Vehicles Volume estimation. Compound error is calculated according to the following formula⁸:

$$\text{total pct error} = \sqrt{(\text{veh volume pct error})^2 + (\text{veh classification pct error})^2}$$

The resulting benchmarks for Truck Volume, by road size bin, are shown in Table 3. Note there is no standard industry guidance on accuracy for very low-volume roads of under 500 in average daily traffic. Also, when reporting Truck Volume errors, road size bins are determined by the *vehicle class* volume rather than total vehicle volume, because this allows for more visibility into how the model performs across different Truck Volumes.

⁵ Krile, R., Todt, F., Schroeder, J. (2015). Assessing Roadway Traffic Count Duration and Frequency Impacts on Annual Average Daily Traffic Estimation (FHWA-PL-16-008). United States. Federal Highway Administration.

⁶ Clarke, N., Co, S., Schewel, L., Wergin, J., Willoughby, C., Yan, L. (2021). "Guidelines for Obtaining AADT Estimates from Non-Traditional Sources," Non-Traditional Methods to Obtain Annual Average Daily Traffic (AADT)

⁷ FHWA (2013). Traffic Monitoring Guide. United States. Federal Highway Administration.
https://www.fhwa.dot.gov/policyinformation/tmguid/tmg_fhwa_pl_13_015.pdf

⁸ Details on error propagation are here: http://www.met.rdg.ac.uk/~swrhgnrj/combining_errors.pdf

	Temporary count expansion		No count available	
Truck Average Daily Traffic*	MAPE (%)	NRMSE (%)	MAPE (%)	NRMSE (%)
A: 0–499	n/a	n/a	n/a	n/a
B: 500 –1,999	14.1	16.3	51.0	65.8
C: 2000–4,999	14.4	19.9	51.0	65.8
D: 5,000–9,999	13.6	17.1	20.6	28.8
E: 10,000+	13.4	16.3	20.6	28.8

*Road size bins will be determined by average daily traffic of the relevant vehicle class

Table 3: Accuracy benchmarks for StreetLight's Truck Volume Metrics, as derived from typical errors in AADT estimation

Model Performance

The following sections present validation results for 36 months of MADT estimates, from January 2019 to December 2021.

CORRELATION BETWEEN ESTIMATED AND ACTUAL MADT

A high predictive model will produce MADT estimates that are strongly correlated with MADT values measured by permanent counters. The plots in Figure 3 depict the correlation between StreetLight's MADT estimates and MADT values by permanent counters for each vehicle class. For all three classes, the correlation is strong, with Pearson correlation coefficients of 0.99, 0.92, and 0.97 for light-, medium-, and heavy-duty, respectively. While most values fall close to the diagonal line, for medium- and heavy-duty, several counter sites with lower MADT fall above the diagonal, indicating overestimation, while some sites with higher MADT fall below the line, indicating underestimation.

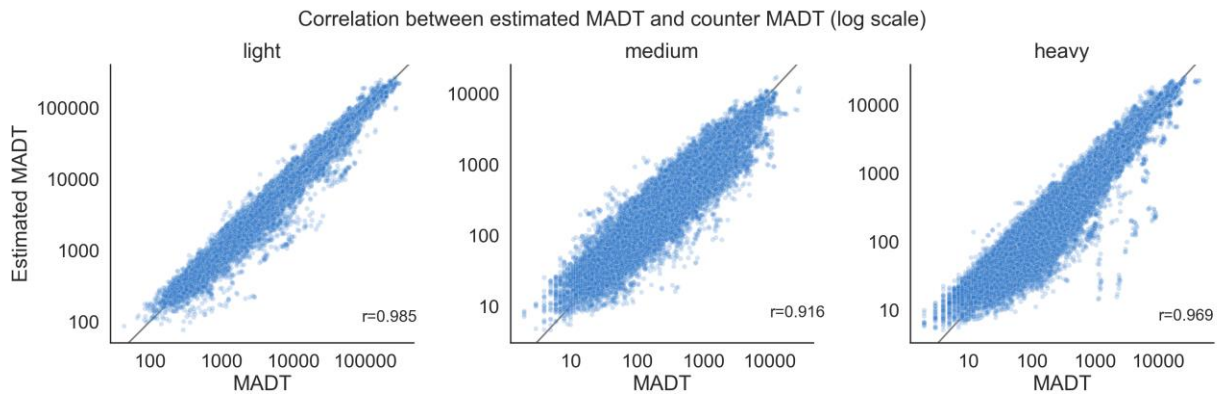


Figure 3: Correlation between StreetLight’s estimated truck MADT and permanent counter MADT, for each vehicle class. (Plots use a log-log scale, where each increment on the axis is a multiple of 10.)

CROSS-VALIDATION RESULTS

Error metrics comparing StreetLight Truck Volume estimates to permanent counter data for MD and HD classes are shown in Table 4 (see appendix Table 4a and 4b for more detail). The results indicate that the HD model is more precise than the MD model, probably for a few reasons. First, we have more HD data points for higher volume roads than we do MD data points. Second, our MD trips sample has fluctuated more than the HD sample has. Third, it’s possible the classification error is greater for MD vehicles, for both permanent counter and GPS data. For example, sensors may have difficulty distinguishing between a four-tire pickup and a six-tire pickup.

For both vehicle classes, error metrics are greater for low-volume roads, which is expected because error metrics in percentage terms will inherently have more variation when the denominator (volume) is smaller. Both MD and HD models have negative median percent error values for roads over 500 MADT (see appendix Table 4a and 4b), which means they tend to underestimate volumes. This may be because most counters have less than 500 daily vehicles for MD and HD, so the model is better at predicting those volumes compared to higher volumes. The constraint of all vehicle class volumes adding up to All Vehicles Volume may also contribute to the negative bias.

For the MD model, we had very few counters with volumes of over 10,000, so it is expected that error metrics are higher for this road size bin.

Table 4 compares the errors for HD and MD models to our benchmarks. For most road sizes, the HD model errors are within five percentage points of temporary count expansion benchmarks. The StreetLight Truck Volume models outperform benchmarks in both vehicle categories when no counts are available for roads with volumes under 5,000. Errors for MD volume are higher than the benchmarks for the large road size category, where training data was scarce. Overall, StreetLight Truck Volume Metrics can be expected to perform about as

well as, or better than, alternative methods of volume estimation. One caveat is that, for MD vehicles on roads with MD volumes above 10,000, we have limited ability to validate model performance due to counter availability.

	MAPE (%)				NRMSE (%)			
	Benchmarks		StreetLight Truck Volume		Benchmarks		StreetLight Truck Volume	
Average Daily Traffic*	Temporary count expansion	No count available	HD	MD	Temporary count expansion	No count available	HD	MD
A: 0–499	n/a	n/a	39.2	36.9	n/a	n/a	48.7	48.7
B: 500–1,999	14.1	51.0	22.8	24.2	16.3	65.8	31.0	34.6
C: 2000–4,999	14.4	51.0	15.9	22.4	19.9	65.8	22.5	30.9
D: 5,000–9,999	13.6	20.6	12.8	23.4	17.1	28.8	19.0	32.5
E: 10,000+	13.4	20.6	14.8	44.0	16.3	28.8	23.8	56.9

*Road size bins are determined by average daily traffic of the relevant vehicle types. For all-vehicles AADT, bins are defined by the total vehicle count, whereas for Truck Volume, bins are based on the specific vehicle class, i.e., MD or HD.

Table 4: StreetLight Truck Volume Error Metrics compared to benchmarks.

For users interested in analyzing medium- and heavy-duty vehicles together, Table 5 shows error metrics for the vehicle categories combined. In general, errors are lower when the vehicle categories are combined.

MD+HD Truck MADT	Count*	MAPE (%)	NRMSE (%)	Median Percent Error (%)	68th Percentile Abs Error	95th Percentile Abs Error
A: 0–499	31,400	31.3	38.4	6.6	34.1	93.7
B: 500–1,999	28,174	20.6	28.2	-3.3	24.7	53.4
C: 2,000 –4,999	13,574	16.8	22.9	-2.3	19.4	45.6
D: 5,000 –9,999	7,956	13.3	18.8	-3.7	14.8	38.1
E: 10,000+	3,116	13.3	23.2	-7.3	14.5	41.4

*Count is total number of data points. Each counter site may have data for multiple months and multiple travel directions.

Table 5: StreetLight Truck Volume error metrics for HD and MD categories combined

TEMPORAL VARIATION IN ERROR METRICS

To check for temporal variation in error metrics, the plots in Figures 4 and 5 show MADT over time for MD and HD vehicles. For MD vehicles, error metrics are stable in 2019 and 2020, then increase in 2021, particularly on low-volume roads. This increase in error in 2021 coincides with

a reduction in our GPS trip sample for MD vehicles—we can assume the weaker signal from GPS trips resulted in less precise estimates. For HD vehicles, error metrics are consistent across time. The variation in error for HD vehicles on high-volume roads (over 10,000 MADT) is due to a small number of counter sites in this road size bin.

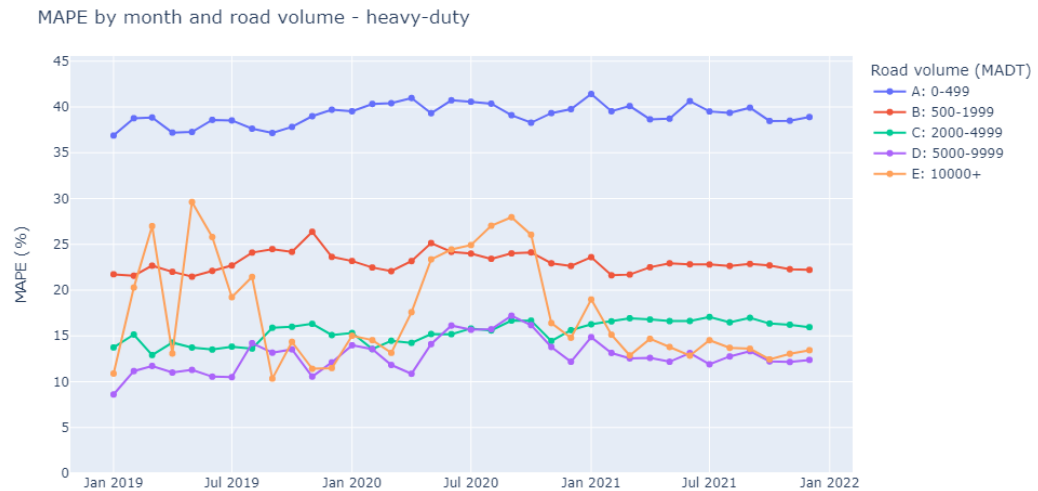


Figure 4: MAPE over time for medium-duty vehicle volume. Increased error in 2021 coincides with a significant reduction in our medium-duty trip sample. (Note road size E: 10,000+ is not shown due to small sample size.)

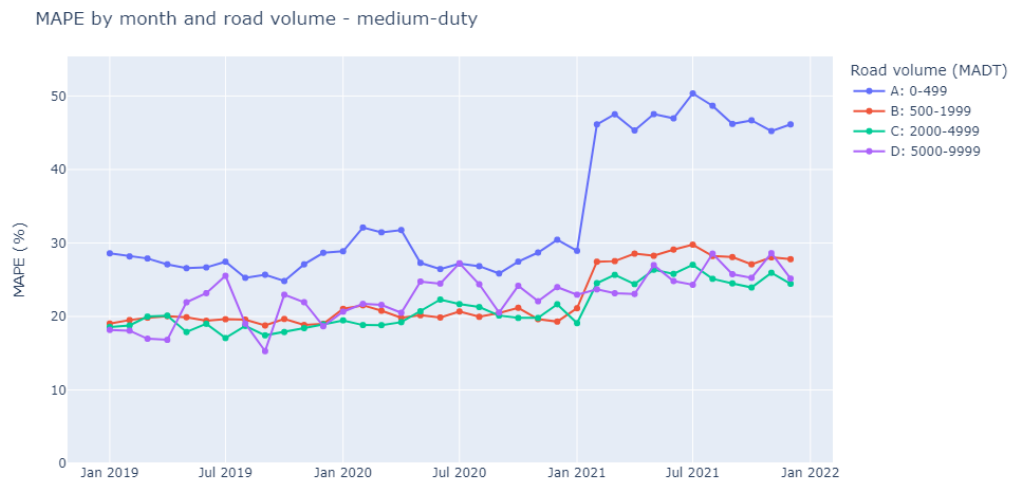


Figure 5: MAPE over time for heavy-duty vehicle volume.

SPATIAL VARIATION IN ERROR METRICS

No notable spatial patterns are observed in the error. For MD, errors are similar across regions. Errors for HD vehicles are similar across regions.

VOLUME TIME TRENDS FOR EXAMPLE ROADS

The plots in Figures 6 and 7 below show StreetLight estimated Truck Volume and permanent counter volume for randomly selected counters, from 2019 to 2021. We can see that time trends in estimated volume generally match trends in permanent counter volume. However, MD estimates for the counters in the C and E road size bins show more deviation from actual volume starting in 2021, which, as noted above, is likely due to decreased GPS sample size at that time.

Estimated MADT and permanent counter MADT for randomly selected sites of varying MD volumes - MD vehicles

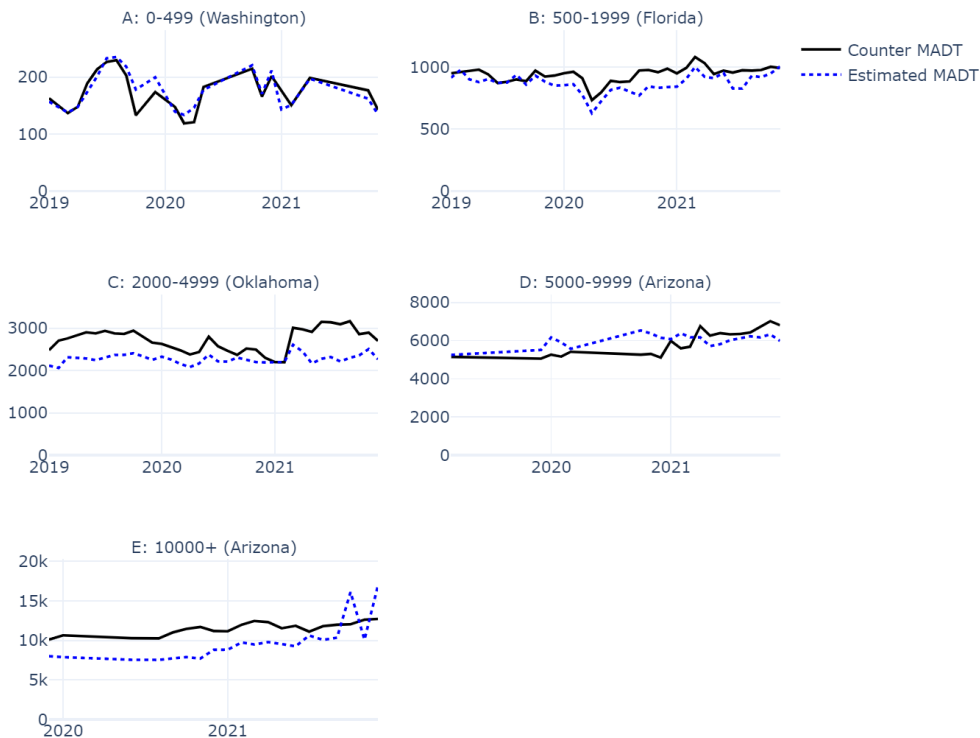


Figure 6: Time trends for sample counter locations, medium-duty vehicles

Estimated MADT and permanent counter MADT for randomly selected sites of varying HD volumes - HD vehicles

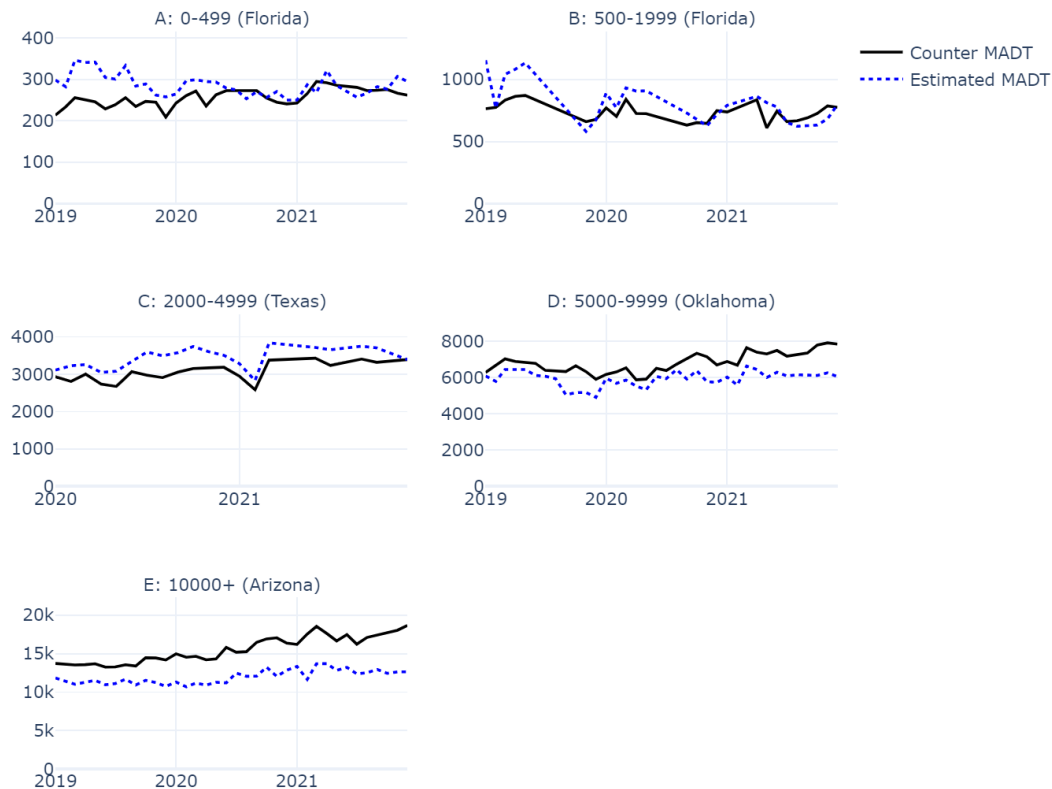


Figure 7: Time trends for sample counter locations, heavy-duty vehicles

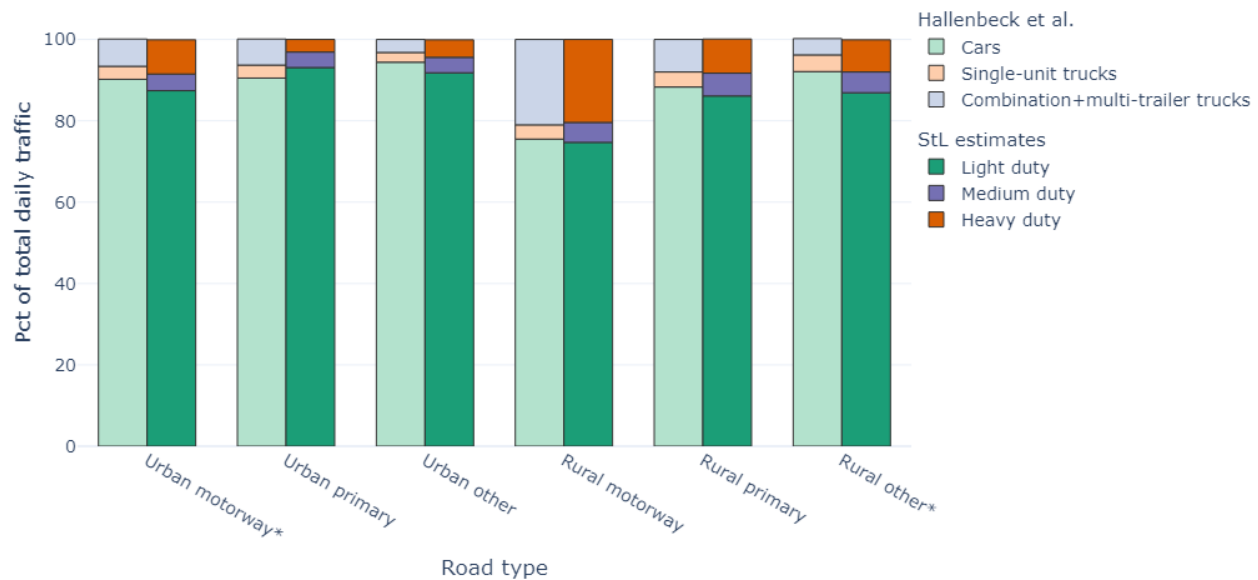
COMPARING STREETLIGHT TRUCK VOLUME WITH EXTERNAL DATA

To further validate the models, we compared our aggregated volume statistics to an external data source other than the permanent counters used in training and validation. The external data came from a study, Hallenbeck et al. (1997), that analyzed volume distributions by vehicle classification.⁹ The study used data from 99 permanent traffic counters across the U.S. whose distribution across space was comparable to those used in our models. Figure 8 shows vehicle class average daily traffic as a percentage of total daily traffic, averaging across the U.S., and compares estimates from our Truck Volume model with those in Hallenbeck et al.'s study. While the study used vehicle and roadway classifications slightly different from those in our Volume Metric—for example, our medium-duty category includes buses and not just trucks—it is still a

⁹ Hallenbeck, M., Rice, M., Smith, B., Cornell-Martinez, C., Wilkinson, J. (1997). Vehicle Volume Distributions By Classification (FHWA-PL-97-025). United States. Federal Highway Administration. https://depts.washington.edu/trac/bulkdisk/pdf/VVD_CLASS.pdf

useful comparison. The study is nearly two decades old, but it is reasonable to assume the U.S.-wide distribution of traffic by vehicle type did not change dramatically in that time period.

Vehicle class average daily traffic as a percentage of total daily traffic



*To make road type categories comparable, Hallenbeck et al.'s results for some road types are combined using averages.

Figure 8: Vehicle class volume as a percent of total volume, comparing StreetLight Truck Volume estimates to external data from Hallenbeck et al.¹⁰

As shown in Figure 8, the StreetLight estimates produce results that are fairly similar to the external data. On rural motorways, for example, Hallenbeck et al. reported 21.0% of traffic to be combination and multi-trailer trucks, while our models estimate 20.4% of traffic as heavy-duty vehicles. In some cases, the values differ: on urban primary roads, our model appears to underestimate HD volume slightly, although Hallenbeck et al.'s sample size for this category was small (only 9 counters).

We also compare StreetLight Truck Volume estimates as a percent of total volume to permanent counter data, as shown in Figure 9. In this case, the estimated vehicle class percentages are nearly identical to those reported by counters. Overall, both of these comparisons add confidence in the validity of the Truck Volume estimates.

¹⁰ Hallenbeck, M., Rice, M., Smith, B., Cornell-Martinez, C., Wilkinson, J. (1997). Vehicle Volume Distributions By Classification (FHWA-PL-97-025). United States. Federal Highway Administration, p. 27. https://depts.washington.edu/trac/bulkdisk/pdf/VVD_CLASS.pdf

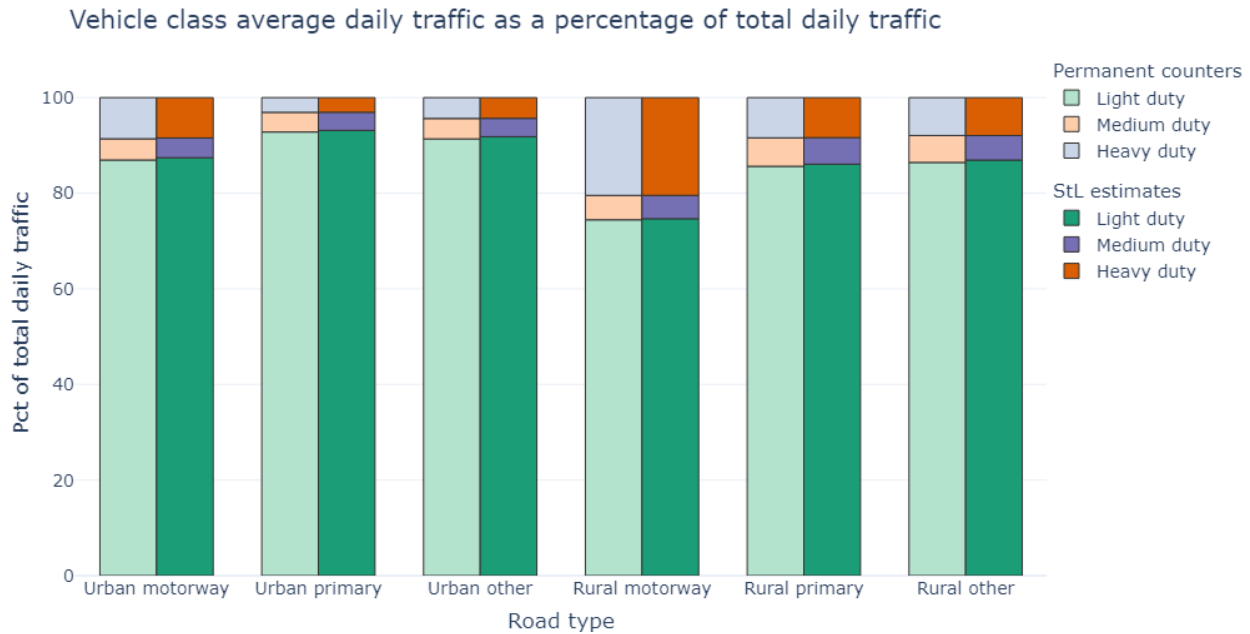


Figure 9: Vehicle class volume as a percent of total volume, comparing StreetLight Truck Volume estimates to permanent counter data.

Limitations

Based on the validation analyses, StreetLight's Truck Volume Metrics are expected to perform comparably to, or better than, alternative means of volume estimation. The Metrics do have some limitations, however. First, MD volume estimates starting in 2021 may be less reliable due to a weaker signal from our trip sample. Second, because our dataset had few permanent counters on roads with more than 10,000 daily MD vehicles, we have limited ability to validate MD Volume Metrics for those roads. Third, both the MD and HD Metrics tend to underestimate Volume for roads that have on average more than 500 vehicles of that class. The underestimation is because most of our training data was for lower volume roads. Finally, because volume estimation for specific days of week and times of day is based on all-vehicles trip counts rather than truck trip counts, these Metrics may be less accurate than monthly volume metrics.

Next Steps

StreetLight's Truck Volume Metrics upgrade StreetLight's Truck Index to closely represent actual vehicle volumes. We are continually working to improve our Metrics by incorporating the most up-to-date data and modeling techniques. In future StreetLight InSight® releases, as new data and methods become available, StreetLight may choose to update the Truck Volume model.

About StreetLight

StreetLight Data, Inc. (“StreetLight”) pioneered the use of Big Data analytics to shed light on how people, goods, and services move, empowering smarter, data-driven transportation decisions. The company applies proprietary machine-learning algorithms and its vast data processing resources to measure travel patterns of vehicles, bicycles and pedestrians, accessible as analytics on the StreetLight InSight® SaaS platform. Acquired by Jacobs as a subsidiary in February 2022, StreetLight provides innovative digital solutions to help communities reduce congestion, improve safe and equitable transportation, and maximize the positive impact of infrastructure investment.

Appendix

Box 1: Error metrics

- **Mean absolute percentage error (MAPE)** is a standard measure of error for predictive machine-learning models. It describes errors well on small roads and treats errors of all sizes equally. The equation for mean absolute percentage error (MAPE) is:

$$MAPE = 100 * \left(\frac{1}{n} \right) * \sum_{i=1}^n \left| \frac{MADT_{Estimate(i)} - MADT_{Permanent\ counter(i)}}{MADT_{Permanent\ counter(i)}} \right|$$

- **Normalized root mean square error (NRMSE)**, another standard error metric in machine learning, penalizes large errors more, making it more sensitive to the accuracy of volume estimation on high-volume roads. The equation for normalized root mean square error (NRMSE) is:

$$NRMSE = 100 * \frac{\sqrt{\left(\frac{1}{n} \right) * \sum_{i=1}^n (MADT_{Permanent\ counter(i)} - MADT_{Estimate(i)})^2}}{\left(\frac{1}{n} * \sum_{i=1}^n (MADT_{Permanent\ counter(i)}) \right)}$$

- **68th percentile absolute percentage error** represents one standard deviation from the mean within a standard bell curve, and thus describes the typical percentage error across the road segments. The 68th percentile absolute percentage error can be interpreted as the expected “typical” error.
- **95th percentile absolute percentage error** represents the upper limit of the expected absolute percentage error.
- **Median percentage error** is a measure of bias in our model, which also indicates how accurate an aggregate value (such as the sum of MADT on all roads in this bin) will be. Values close to 0 suggest that our model has low bias, while positive values indicate overestimation and negative values indicate underestimation.

State	Number of sites*	Number of counters	State	Number of sites*	Number of counters
Alabama	14	28	Mississippi	97	192
Arizona	139	269	Missouri	67	67
Arkansas	41	81	Montana	63	119
California	73	134	Nevada	15	29
Colorado	52	100	New Jersey	73	122
Connecticut	10	17	New Mexico	16	32
Delaware	18	32	New York	81	157
District of Columbia	2	2	North Dakota	60	117
Florida	288	566	Ohio	75	144
Georgia	197	387	Oklahoma	92	182
Idaho	35	67	Pennsylvania	112	219
Illinois	68	131	Rhode Island	16	30
Indiana	81	158	South Carolina	62	117
Iowa	64	124	South Dakota	15	29
Kansas	15	29	Texas	131	254
Kentucky	49	91	Vermont	11	22
Maine	16	28	Virginia	243	396
Maryland	26	50	Washington	159	290
Massachusetts	81	150	West Virginia	45	81
Michigan	53	106	Wisconsin	85	169
Minnesota	60	114	Wyoming	102	198
			Total	3,002	5,630

*Each site may have counters for multiple travel directions.

Table A1: Number of unique permanent counters and sites by state

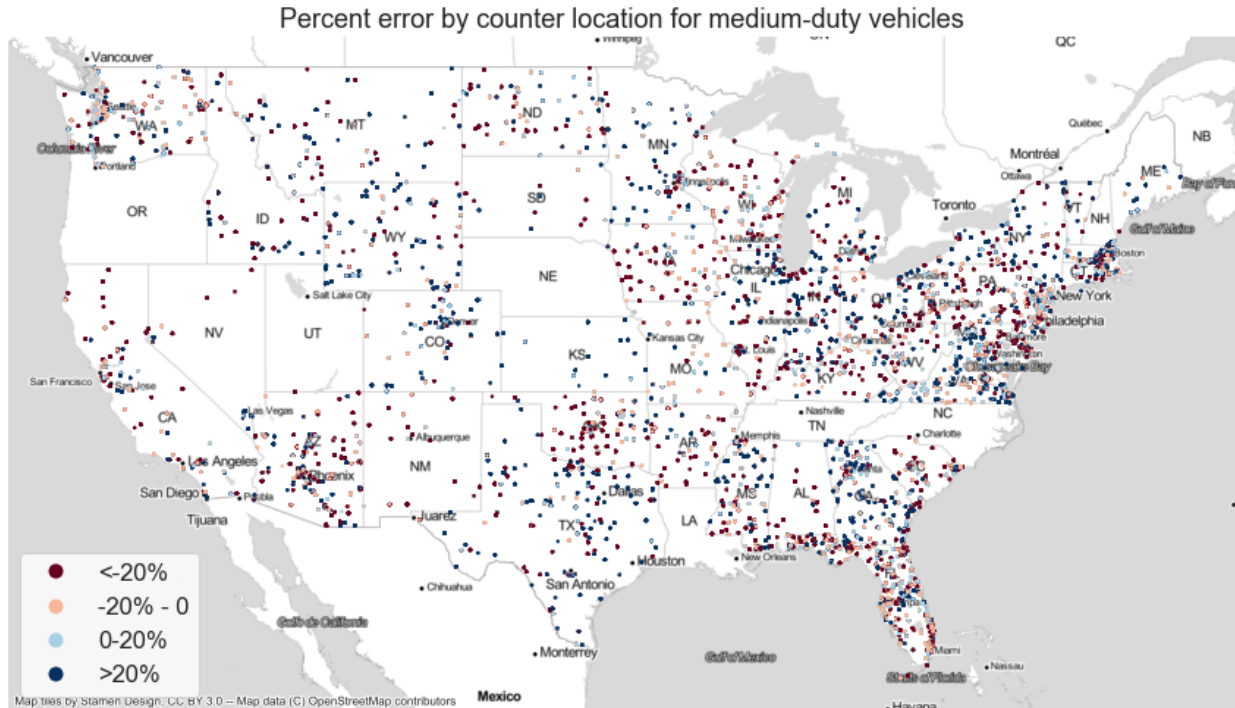


Figure A1: Percentage error between StreetLight estimated MADT and permanent counter MADT by location for MD vehicles.

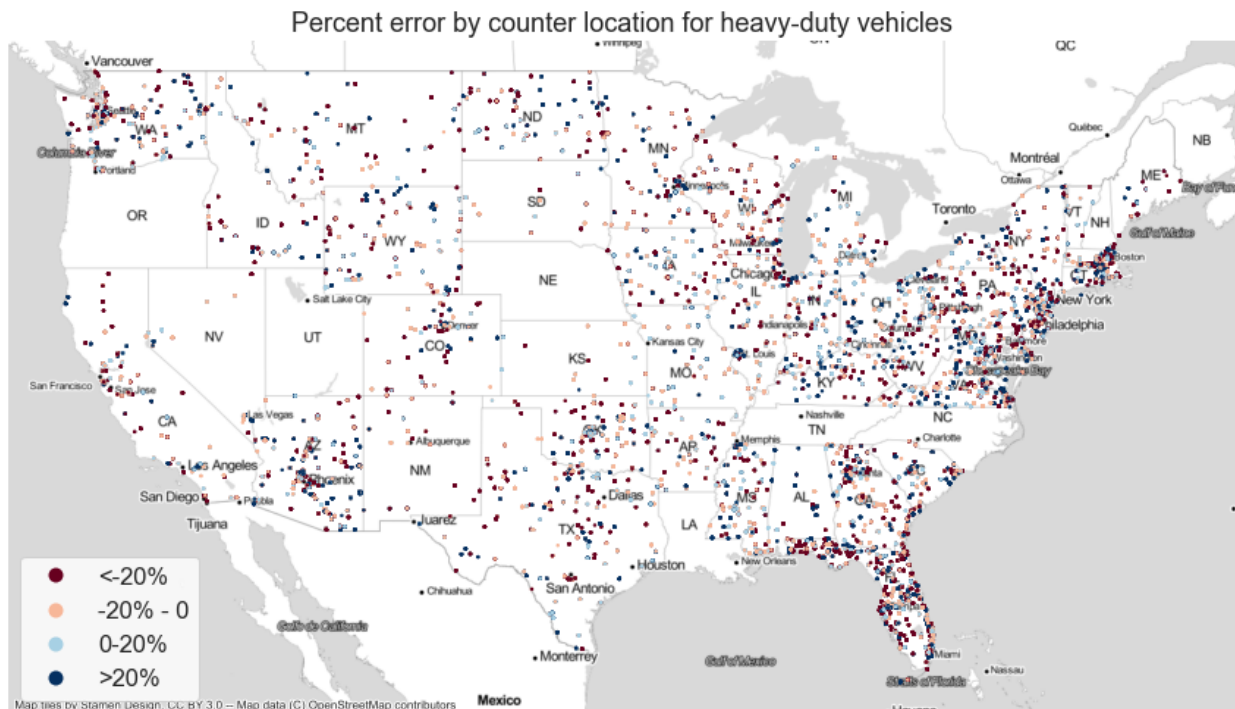


Figure A2: Percentage error between StreetLight estimated MADT and permanent counter MADT by location for HD vehicles.

MD Truck Average Daily Traffic	Count*	MAPE (%)	NRMSE (%)	Median Percent Error (%)	68th Percentile Abs Error	95th Percentile Abs Error
A: 0-499	49,757	36.9	48.7	4.6	37.3	116.6
B: 500-1,999	26,123	24.2	34.6	-4.3	28.1	64.7
C: 2,000-4,999	7,059	22.4	30.9	-10.3	26.9	63.5
D: 5,000 - 9,999	1,190	23.4	32.5	-17.3	25.6	71.2
E: 10,000+	91	44.0	56.9	-42.0	54.6	87.0

*Count is total number of data points. Each counter site may have data for multiple months and multiple travel directions.

Table 4a: Medium-Duty Vehicle Volume Error Metrics

HD Truck Average Daily Traffic	Count*	MAPE (%)	NRMSE (%)	Median Percent Error (%)	68th Percentile Abs Error	95th Percentile Abs Error
A: 0 -499	47,807	39.2	48.7	5.8	43.0	120.4
B: 500 - 1,999	19,298	22.8	31.0	-3.9	26.9	60.7
C: 2,000 - 4,999	10,865	15.9	22.5	-2.3	17.4	46.1
D: 5,000 - 9,999	4,958	12.8	19.0	-3.7	14.3	33.4
E: 10,000+	1,292	14.8	23.8	-8.4	15.8	48.2

*Count is total number of data points. Each counter site may have data for multiple months and multiple travel directions.

Table 4b: Heavy-Duty Vehicle Volume Error Metrics



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